

Mayank Aggarwal

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<https://github.com/mayankaggarwal>

Experience

Principal Software Engineer, S&P Global (June 2023 – Present)

Leading Team to deliver various migration projects like incorporating AI capabilities in product, migrating legacy C++ components to C# .net based solution, migration of Active X based pages to Angular 15, Integrating Web API with OIDC provider and exposing API to public domain and migration of on-prem system to AWS using Lift and shift technique.

Software Development Engineer 3, ShipBob (July 2022 – June 2023)

Developing software systems which help in shipment of Products across the globe by integrating third party carrier aggregators. Contributed in incorporating reliability in the systems, improving monitoring using Azure App Insights and reducing time to deliver by improving pipelines.

Senior Software Engineer, IHS Markit (April 2020 – July 2022)

Developing services which could be used across multiple teams for identity management, tenant management, permissions, and configurations. Majorly involved in leveraging Kubernetes specific features in .Net based applications.

Lead Developer, Dunnhumby (April 2017 – April 2020)

Developing new components and services for the product. Migrating product on newer tech stack. Implementing and incorporating client requirements in the product.

System Specialist, Orange Business Services (December 2014 – April 2017)

Design and Develop Data Warehouse and lead the team providing architectural guidance. Maintaining existing applications.

Senior Systems Engineer, Infosys Limited (February 2012 – October 2014)

Educational Background

Formal Education

Bachelor of Technology, Uttarakhand Technical University [2007-2011] with **75.6%**

Senior Secondary (XII), CBSE Board [2007] with **77.8%**

Secondary (X), CBSE Board [2005] with **80.4%**

Skills

Programming

C#, TypeScript, SQL

Software's

MS SQL Server 2022, Copilot, Ollama, MySQL, Visual Studio 2026, Kubernetes, gitlab pipelines, Azure DevOps, PostgreSQL 9.6, MS SQL Server 2014, Elastic Search 7.x, Aerospike, Couchbase, Docker, ETL tools like Talend and Pentaho, and MS Office.

Frameworks/SDK

.Net 8, Entity Framework, Ollamasharp, OpenAPI SDK, Microsoft.Extensions.AI package, MCP framework, Identity Server 4, Angular 15, SignalR, RabbitMQ, gRPC and helm charts

Operating System

Windows (11 and below), Windows Server and Linux (RHEL 6.6).

Cloud

Leveraged services pertaining to cloud providers like AWS (EKS, ECS, Fargate, Secrets manager, S3, AWS hosted RabbitMQ etc.) and Azure (Cosmos DB, WebApps, Functions, EventGrid, ServiceBus)

Projects

Equities Team (Current)

Involved in delivering various migration projects for the product developed for Primary Markets.

- Currently Migrating Classic ASP based pages to Angular based application.
- Enabled development team to use **custom AI agents and skills enabled using markdown files** to be used with **copilot** for identifying vulnerabilities and probable fix, create unit tests, provision default applications with all dependencies and integrations, knowledge document creation etc. Added instruction file to be used by **Copilot code review agent** in multiple github repos.
- Delivered project which exposed product's API capabilities over **MCP (Model Context Protocol) to be consumed by clients AI agents**.
- Delivered massive project to migrate 20-year-old legacy C++ codebase to modern .Net based solution where migrated certain components individual and led team of 10-15 developers. Involved in **design, project planning, sprint planning, implementation, development and release**.
- Delivered project related to Integrating **Web API with OIDC provider** and exposing API to public domain and Migrated product from **on-prem infrastructure to AWS using Lift and shift technique**.
- Lead team involved in migration of Active X based pages to Angular based application using Angular 15, Akita for state management etc.

Transportation Team (At ShipBob)

Involved in building services around **Supply Chain Management** by enabling purchasing labels for Orders by integrating with Carrier Partners using services developed **with Dotnet Core 3.1, Azure Functions, Azure Service Bus and Microsoft SQL Server**. Contributed by incorporating Circuit Breaker Pattern, Azure Alerts for better monitoring and helped in decoupling of existing monolith to Microservices. Worked on developing address validation service in order to make deliveries of products to verified addresses.

Enterprise Data Management (At IHS Markit)

Involved as part of the Platform services team for building common services related to **Authentication and Authorization**, Configuration and **Tenant management** leveraging Angular 11, Dotnet Core 3.1, gitlab pipelines, docker, Kubernetes and AWS.

Worked on leveraging Kubernetes based features like **Secret Management, Certificate Management, Health Checks** in .net based application.

Worked on developing service for **RBAC management using Casbin**.

Worked on incorporating **SSO using Identity Server 4** by integrating with ADFS and Okta identity provider.

Campaign Manager Portal (At Dunnhumby)

Involved in developing the Front-end application using **Angular 5, DotNet Core, ASP.Net Core Web API, Elastic Search, gRPC, Docker, Google Cloud Platform, Google Cloud Storage and PostgreSQL**

Product Recommendation System (At Dunnhumby)

Involved in maintenance and deployments of product to a new market using **ASP.Net Web API, Specflow, T4 templates, Teamcity**

Group Identity Data Warehouse for Orange Corporate HR (At OBS)

Involved in creating **the technical architecture** of the project, developing database components using **MySQL**, conceptualizing **Talend Jobs**, leading a team of 4 members handling tasks related to Talend and Cognos.

Group Identity Referential for Orange Corporate HR (At OBS)

Development of **WCF services** and maintenance of **ASP.Net based web applications**. Fetching requirements from **the French client** and preparing technical specifications.

Analytics Platform as part of TransOrg Analytics

Migration of the Churn Prediction framework from Alterian-KXEN based solution to Pentaho, PostgreSQL and R based solution. Worked specifically on developing **ETL Jobs** and **database Schema development**.

Data Entry and Enrichment Platform as part of Infosys Labs

Automating the Data entry process of invoices into SAP systems with minimum human intervention. Worked on developing components related to applying OCR's efficiently to read data, enriching the **OCR's** results using text analytics, developing **WCF REST based web services**, windows services, and **ASP.Net based Website**

PARE as part of Infosys Labs

Automating the processing of report creation right from download from SAP systems to transformation. Involved in developing **ASP.Net based Web applications** for scheduling automation jobs.

Design and Developing Effective ways of Customer Segmentation as part of Microsoft Technology Center (MTC)

Learning the interest of customers and identifying patterns using their browsing or buying products at an e-commerce site using IIS logs. Involved in **development of framework** for analyzing and identifying patterns using **machine learning algorithms and MongoDB**

Developing Context Aware Collaboration platform as part of MTC

Virtual help desk for organizations applications, learning through users working patterns on their workstations, obtaining the context of the knowledge workers current work etc. Involved in the development of a framework for analyzing and identifying patterns using **machine learning algorithms**, developing **windows 8 apps**, and creating **WCF services**.

Designing and Developing solutions using Microsoft Lync as part of

MTC Microsoft Lync, Windows 8 apps, WCF.

Enabling Microsoft Lync capabilities in custom windows 8 and Windows phone 8 applications.

POC's of the Windows 8 app for MNC's as part of MTC

HTML 5.0, CSS3.0, JavaScript and WinJS Library.

Proof of Concept for an MNC was developed for mobility field service on the Windows 8 platform.

Infosys Library Management System as a training project

ASP.Net, ADO.Net and SQL Server 2008.

The Library Management System was developed with a team of 7.

Achievements

White paper published for SRII 2014 (San Jose) conference.

Guinness World Records, record holder for "Most participants in an App Development Marathon" at Windows AppFest 2012 Bangalore.

Memory Champ App selected for Windows Store at Windows AppFest 2012.

Completed Infosys Limited entry level training in the Microsoft track with a CGPA of 4.93 out of 5.

Accomplished graduation with **Honors**.

Won '**CHITRAKALA-Kalabhoshan' Award** in National Drawing and Writing Competition conducted by Bharat Manav Samaj Vikas Kendra (Dadar-Mumbai).

(Mayank Aggarwal)